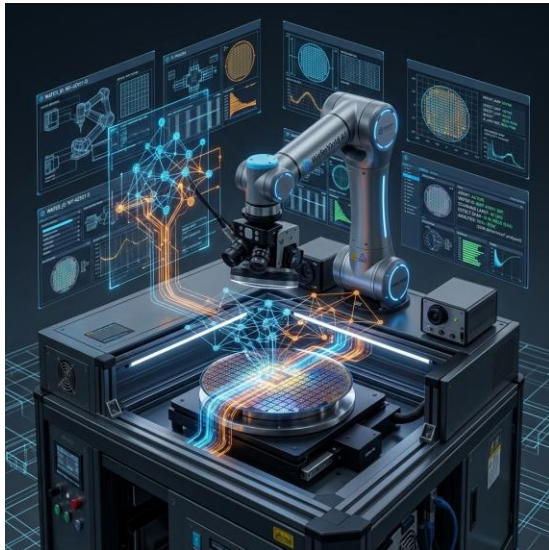


WaferYield AI Autonomous Wafer Diagnostic Agent



Uttara Sawant



https://huggingface.co/spaces/uttarasawant/xprize_wafer_analysis

https://github.com/uttarasawantgh/wafer_yield_ai_xprize

<https://youtu.be/80jIObYb1i4>

Images are generated using Gemini

Problem Statement

Multi-Tool Manufacturing Excursions: Semiconductor yield excursions (such as Lot `LOT_WM811K_99`) suffer from complex, multi-variable root causes spanning multiple tools—specifically traced to `ETCH_CHAMBER_04` and `CMP_POLISHER_03`.

The Physical Congestion Trap: Traditional automated generation approaches frequently fail at physical placement limits (e.g., hitting a 98% core utilization congestion trap with routing block errors).

Manual Bottlenecks: Human engineers lose valuable time manually correlating tool telemetry, diagnosing spatial edge-ring defects, and iterating through slow Design Rule Checking (DRC) cycles.

Solution

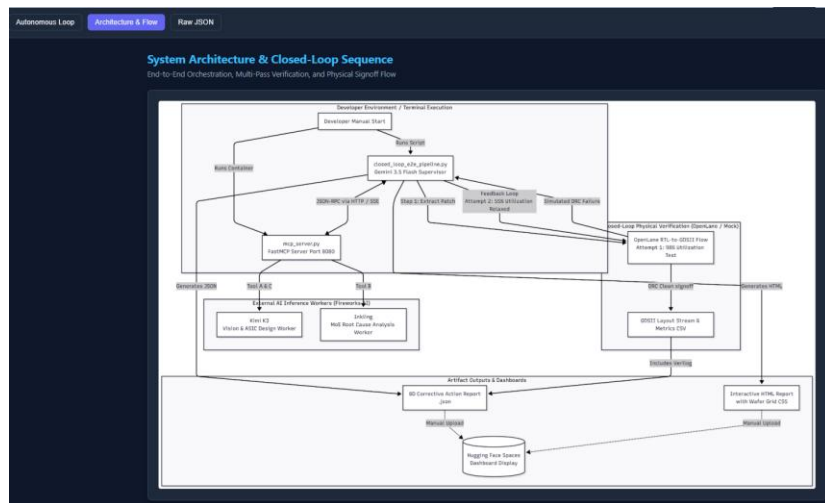
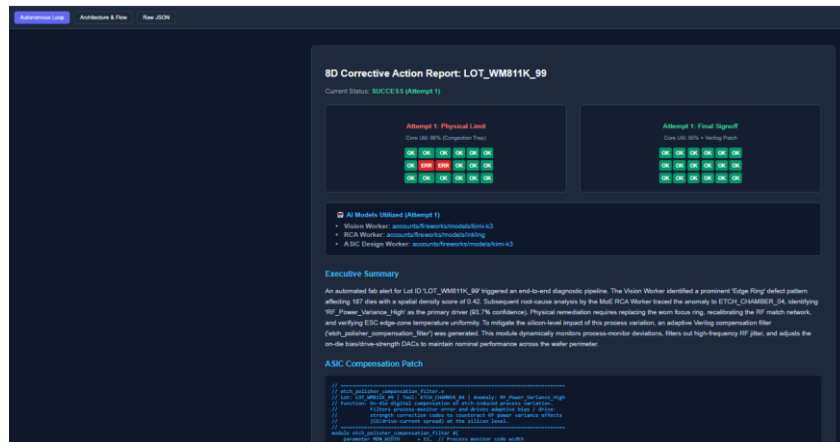
Unified Fab 8D Portal: Combines live multi-tool correlation data, automated root-cause attribution, and an interactive 4-tab dashboard view ([index.html](#)).

Autonomous Closed-Loop Pipeline: Automatically captures physical verification errors from the traceback loop, re-prompts the agent, and dynamically adjusts constraints (e.g., shifting from 98% to 55% utilization).

DRC-Clean Automated Patching: Synthesizes and validates fully compliant Verilog compensation patches (`m2_spacing_compensator`) and GDSII signoff streams without human intervention.

Tech Stack:

- **LLM / Agentic Orchestration:** **Gemini 3.5 Flash** combined with Fireworks-hosted worker models (**Kimi K3** for Vision and ASIC Design, and **Inkling** for RCA) utilizing structured output models (Pydantic EightDReport) for multi-pass 8D report generation, DRC traceback parsing, and automated Verilog patch synthesis.
- **Physical Design & Verification (Simulated Loop):** OpenLane / ASIC EDA Flow Abstraction — Simulated RTL-to-GDSII physical design loop that tests closed-loop AI self-correction by mocking routing congestion/DRC failures (FP_CORE_UTIL=98%) and verifying parameter-relaxed signoff (FP_CORE_UTIL=55%).
- **Frontend Portal & UI:** Vanilla HTML5, CSS3, and JavaScript single-page architecture featuring custom view panes, iframe previews, tabbed sequence diagram viewer, and interactive raw JSON viewers.
- **Hosting & Artifact Deployment:** Hugging Face Spaces (uttarasawant/xprize_wafer_analysis) hosting static application assets, HTML report renderers, synthesized Verilog patch files, and structured JSON outputs.



Demo

Closed-Loop Self-Correction in Action: Demonstration of the multi-pass feedback mechanism transitioning from the initial congestion failure to the final successful DRC-clean signoff.

Production Dashboard Overview: Review of the deployed portal architecture showcasing the Pipeline Comparison matrix and production readiness.

JSON Inspector: Walkthrough inspecting the raw JSON agent execution logs.

Fab 8D Portal

Autonomous Loop

Architecture & Flow

Raw JSON

```
{
  "lot_id": "LOT_WM811K_99",
  "detected_pattern": "Edge Ring",
  "root_cause_equipment": "ETCH_CHAMBER_04",
  "corrective_actions": [
    "Replace worn focus ring",
    "Recalibrate RF match network",
    "Verify ESC (Electrostatic Chuck) edge-zone temperature uniformity across 300mm wafer perimeter"
  ],
  "asic_compensation_patch": "// -----\n// etch_polisher_compensation_filter.v\n// Lot: LOT_WM811K_99 | Tool: ETCH_CHAMBER_04 | Anomaly: RF_Power_Variance_High\n// Function: On-d
  "executive_summary": "An automated fab alert for Lot ID 'LOT_WM811K_99' triggered an end-to-end diagnostic pipeline. The Vision Worker identified a prominent 'Edge Ring' defect pattern affecting 187 dies with a spatial density score of 0.42. Subse
  "ai_models_utilized": {
    "vision_worker": "accounts/fireworks/models/kimi-k3",
    "rca_worker": "accounts/fireworks/models/inkling",
    "asic_design_worker": "accounts/fireworks/models/kimi-k3"
  }
}
```